VNUHCM-UNIVERSITY OF SCIENCE
FACULTY OF INFORMATION TECHNOLOGY

COURSE SYLLABUS
CS333 – Introduction to Operating Systems

1. GENERAL INFORMATION

Course name:	**Introduction to Operating Systems**
Course name (in Vietnamese):	Nhập môn Hệ điều hành
Course ID:	CS333
Knowledge block:	Chuyên ngành
Number of credits:	4
Credit hours for theory:	40
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	Nonce
Prior-course:	Computer Systems
Instructors:	Trần Trung Dũng

2. COURSE DESCRIPTION

This course covers the important problems in operating system design and implementation. It will also cover the tradeoffs that can be made between performance and functionality during the design and implementation of an operating system. Particular emphasis will be given to three major OS subsystems: process management (processes, threads, CPU scheduling, synchronization, and deadlock), memory management (segmentation, paging, swapping), and file systems; and on operating system support for distributed systems.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Learn a lot of practical information about how programming languages, operating systems, and architectures interact and how to use each effectively.	5.1.1, 5.1.3, 5.2.1, 5.2.2, 5.3.1, 6.1.1
G2	Learn about how concurrency and distributed systems communicate and work correctly	5.1.1, 5.1.3, 5.2.1

G3	Have knowledge to more effectively use and manipulate computers and computer program	5.1.3, 5.2.1, 5.2.2, 5.3.1
G4	Team work & Independent thinking	2.2, 2.3.1
G5	Understanding keywords	2.4.3, 2.4.5

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Explain the fundamental components of a computer operating system	T
G1.2	Discuss and explain the policies for scheduling, deadlocks, memory management, synchronization, system calls, and file systems	T
G2.1	Extrapolate the interactions among the various components of computing systems	T
G2.2	Construct the following OS components: System calls, Schedulers, Memory management systems, Virtual Memory and Paging systems	TU
G3.1	Illustrate, construct, compose and design solutions via C/C++ programs, and through XV6	TU
G4.1	Criticize any topic related to OS	TU
G5.1	Show the ability of working independently or working in group	TU

5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Overview A brief of History	G1.1	Lecturing Q&A, Group discussion QZ1: Quiz 1

2	Architecture	G1.1	Lecturing Demonstration, Q&A QZ2: Quiz 2 Project 1 hand out
3	Processes and Threads	G2.1, G2.2	Lecturing Demonstration, discussion
4	Synchronnization	G2.1, G2.2	Lecturing Demonstration, discussion Project 2 submitted
5	CPU Schedule	G2.1, G2.2	Question & answer Case study and discussion Project 2 hand out
6	Memory management	G2.1, G2.2	Lecturing Demonstration
7	Virtual memory	G2.1, G2.2	Lecturing Q&A, discussion
8	Disk: HDD & SSD	G2.1, G2.2	Lecturing
9	File systems	G2.1, G2.2	Case study, discussion Demonstration
10	Review	G1.1, G1.2, G2.1, G2.2, G3.1, G4.1	Lecturing Q&A, Discussion Project 2 submitted

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the corresponding topic and ask students to do their lab exercises either on computer in the lab or at home. All the lab work submitted will be graded. There would be a final exam for lab work.

6. LAB PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Project 1 Intro xv6	G1.1, G5.1	Lecturing Q&A, Group discussion
2	Project 2 System call	G1.1, G5.1	Lecturing, Demonstration, Q&A

3	Project 3 Pagetable	G1.1, G5.1	Lecturing, Demonstration, Q&A

7. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	**Assignments**			**10%**
A11	Quizzes: QZ1, QZ2, QZ3, and QZ4.	Small quizzes in class for each topic	G1.1, G3.1	5%
A12	Homework: HW1, HW2, and HW3	HW1, HW3: reading comprehension and writing reports in English HW2, HW3: practicing based on knowledge taught in class	G1.1, G3.1	5%
A2	**Projects**			**30%**
A21	Project 1	Intro xv6	G1.1, G5.1	10%
A22	Project 2	System call	G1.1, G5.1	10%
A23	Project 3	Page Table	G1.1, G5.1	10%
A3	**Exams**			**70%**

A32	Midterm exam	Open book exam. Describe the understanding of different topics, analyze & program to solve problems		35%
A33	Final exam	Open book exam. Describe the understanding of different topics, analyze & program to solve problems		35%

8. RESOURCES

Textbooks

- Operating System Concepts 10th Edition, by Abraham Silberschatz, Peter B. Galvin, Greg Gagne.

  Publisher : Wiley; 10th edition (May 2, 2018)
  ISBN-10 : 1119456339
  ISBN-13 : 978-1119456339

Software

Linux Operation System at https://github.com/torvalds/linux

9. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.